

FinRep — Feature Guide: Analytics (Phase 4) & AUP Readiness (Phase 2)

A detailed walkthrough of everything shipped in the Analytics & Insights dashboard and the AUP / Year-End Readiness module

Prepared: June 2026 · **Audience:** product, leadership, and prospective school customers ·

Companion to: `PROJECT_OVERVIEW.md`

How to read this document

This guide documents two completed modules that sit on top of FinRep's proven financial engine:

- **Phase 4 — Analytics & Insights Dashboard** (`/analytics`): turns the saved statements into an at-a-glance, customizable view of a school's financial and operational health.
- **Phase 2 — AUP Readiness & Year-End Review** (`/readiness`): helps a Florida private school prepare for its mandatory scholarship-compliance review (the Agreed-Upon Procedures, or "AUP") and its year-end financial review.

Two design principles run through both modules and are worth stating once:

1. **The logic lives in sealed, testable "engines."** The analytics metrics and the compliance rules are each implemented as a **pure, deterministic library** (`@finrep/analytics` , `@finrep/compliance`) — the same inputs always produce the same outputs, with no hidden state, clock, or randomness. Every metric and rule is unit-tested.
2. **Everything reads from the reproducible snapshot.** Both modules consume the *saved* statement snapshot the engine already produced; they never re-invent or silently re-compute the underlying numbers.

Every screen is permission-aware (Owner / Accountant / Viewer) and tied to an active subscription or live trial. Viewers can always *read*; editing and generation require the right role and an entitled account.

Part I — Phase 4: Analytics & Insights Dashboard

What it is: a per-school dashboard (/analytics) that converts the statements into living operational insight — with health-graded metrics, trends, drill-downs, an AI summary, and a layout each school can tailor. Delivered in four slices (4A–4D).

1. The metric catalog

Metrics come in two tiers. **Financial-only** metrics derive purely from the statements (available immediately). **Operational** metrics also need two inputs that don't live in a trial balance — enrollment and financial-aid totals — entered once per period.

Tier 1 — Financial metrics (no extra input required)

Metric	Formula	Unit
Operating Margin	$(\text{Total revenue} - \text{Total expenses}) \div \text{Total revenue}$	percent
Days Cash on Hand	$\text{Cash} \div (\text{Total expenses} \div 365)$	days
Months of Operating Reserve	$\text{Unrestricted net assets} \div (\text{Total expenses} \div 12)$	months
Tuition Dependency	$\text{Tuition \& fees} \div \text{Total revenue}$	percent
Revenue Mix	$\text{Each revenue category} \div \text{Total revenue}$	share (donut)
Expense Mix	$\text{Each expense category} \div \text{Total expenses}$	share (donut)

Tier 2 — Operational metrics (need enrollment / aid)

Metric	Formula	Unit
Cost per Pupil	$\text{Total expenses} \div \text{Enrollment}$	currency
Net Tuition per Student	$(\text{Gross tuition} - \text{Financial aid}) \div \text{Enrollment}$	currency
Aid per Enrolled Student	$\text{Financial aid} \div \text{Enrollment}$	currency
Aid per Aided Student	$\text{Financial aid} \div \text{Students on aid}$	currency
Tuition Discount Rate	$\text{Financial aid} \div \text{Gross tuition}$	percent
% of Students on Aid	$\text{Students on aid} \div \text{Enrollment}$	percent

Honesty contract: when a metric's inputs are missing (e.g. enrollment not yet entered), the metric reports "**needs data**" with the exact missing input named — never a misleading zero.

2. Health status & targets (4D)

Five ratio metrics are graded **good** / **watch** / **risk** against sensible private-school sector targets (tunable later; defaults shown). The other metrics are **neutral** — shown with their trend, but no universal good/bad coloring.

Metric	Direction	Good	Watch	Risk
Operating Margin	higher better	$\geq 3\%$	0–3%	$< 0\%$
Days Cash on Hand	higher better	≥ 60	30–60	< 30
Months of Operating Reserve	higher better	≥ 6	3–6	< 3
Tuition Dependency	lower better	$\leq 70\%$	70–85%	$> 85\%$
Tuition Discount Rate	lower better	$\leq 20\%$	20–35%	$> 35\%$

3. What each slice delivered

4A — Financial dashboard. The `@finrep/analytics` engine + the Tier-1 metric cards (animated value, period-over-period delta, sparkline), the revenue/expense donuts, and a multi-period trend chart — all computed from the saved snapshot.

4B — Operational data + Tier-2 metrics. A per-period intake for **enrollment** (headcount + optional FTE), **students on aid**, and **financial-aid total**, which lights up the six Tier-2 metrics (cost per pupil, the aid metrics, the discount rate, % on aid). These same inputs feed the Phase-2 compliance checks.

4C — Per-school customization. An Owner-only **Customize mode**: show/hide metrics, drag to reorder, choose a chart variant, and set card width — saved per school and applied for everyone. When nothing is customized, a sensible default layout is shown.

4D — Redesign & intelligence layer. A cleaner "hero vitals + grouped sections" layout with: - **Hero tiles** for the financial-health vitals, color-graded by status; - a **drill-down drawer** — click any metric to see its full trend, its **formula**, and the exact input values feeding it; - an **AI insight line** — a plain-language summary of what changed and what to watch (e.g. "*Tuition dependency remains high at 87% — the top priority to address*"). It works out of the box with a deterministic, rule-based generator and can be upgraded to a Claude-written narrative by supplying an API key; - **freshness cues** — a context bar showing the active period, "*data as of*" the snapshot date, and how recently it was updated.

Access: analytics is included for every customer on an active subscription or live trial — there is no separate analytics paywall. Owners customize; Accountants and Viewers see the configured layout read-only.

Part II — Phase 2: AUP Readiness & Year-End Review

What it is: a **Review Readiness** workspace (`/readiness`) that mirrors the published Step Up For Students Agreed-Upon-Procedures template and the governing Florida statutes, so a school can self-check **before** its CPA engagement and hand the reviewer a clean packet.

Framing (shown prominently in-app): this is a **readiness pre-flag** — it is **not** the official AUP submission and **not** legal or audit advice. Every rule is grounded in cited statute/template language (see `PHASE_2_AUP_RESEARCH.md`).

1. The trigger and the framework

- **Who must do the AUP:** a participating private school that receives **more than \$250,000** in aggregate scholarship funds (across FTC, FES-EO, FES-UA) in a school year must engage a CPA to perform the AUP and file the report by **September 15**.
- **Six AUP sections** organize every check: **I** School Eligibility · **II** Accounting System · **III** Financial Controls · **IV** Deposit & Classification of Scholarship Funds · **V** Education-Related Expenses · **VI** Tuition / Operating Term / Attendance (plus an **Eligibility** gate).
- **Two severities:** a **Reportable** exception, and a **Material** exception (the same reportable issue three years running) — a Material exception **requires a Corrective Action Plan**, and repeated ones can render the school ineligible.
- **Versioned, not hard-coded:** the rule set is stamped `fl-scholarship-aup v2025.1` (statute year 2024) and pinned by program tier, so a procedure change is a version bump — not a code rewrite.

2. 2A — Exception pre-flagging (the rules engine)

Thirteen checks run against each period. Each carries a **status** (pass / reportable / material / needs-data / manual / not-applicable), a **severity**, a **kind** (auto = from the statements; intake = from a short attestation; checklist = document the CPA will sample), and its **citation**.

Rule	§	What it checks	Severity	Kind	Citation
\$250,000 AUP trigger	V	Scholarship funds received > \$250k → AUP required (due Sep 15)	info	intake	s.1002.421(1)(q)
Trial balance balances	II	Self-balancing system produces a balanced TB	reportable	auto	AUP §II
Education expenses ≥ scholarships	V	Total education expenses ≥ scholarship funds received	reportable	auto	AUP §V
Non-education expenses flagged	V	Athletics / student activities / transportation / food can't count toward coverage	reportable	auto	AUP §V
Funds at insured institution	III	Scholarship funds at an FDIC/ NCUA-insured bank	reportable	intake	AUP §III.A
Bank-rating review > \$250k	III	If avg balance > \$250k, bank rating reviewed (top two)	reportable	intake	AUP §III.A
60-day bank reconciliation	III	Reconciled within 60 days and independently reviewed	material	intake	AUP §III.B
Scholarship deposit tracing	IV	Sample of ACH payments traced to deposits / GL / student accounts	reportable	checklist	AUP §IV
DOE status "approved"	I	DOE compliance/approval status is "approved"	reportable	intake	§I; s. 1002.395(2)(i)
Operated ≥ 3 years or surety bond	ELIG	3+ school years, else a surety bond / LOC posted	gate	intake	s.1002.421(1)(f)1.
FES-UA \$50,000 cap	ELIG	No FES-UA ESA balance over \$50,000 (<i>UA tier only</i>)	reportable	intake	s.1002.394(12)(b)11.
FES-UA dormancy / closure	ELIG	Dormant-account closure rules (<i>UA tier only</i>)	reportable	checklist	s.1002.394(5)(b)3.
Prudential red flags	VI	Negative net assets / cash, operating deficit, low days-cash	watch	auto	prudential

The compliance intake collects the handful of attestation facts that aren't in a trial balance, so most sections produce a real verdict: scholarship funds received, programs, funds-at-insured-institution, avg-balance-over-\$250k, bank-rating-reviewed, reconciled-within-60-days, independently-reviewed, DOE-approved, years-in-operation, surety-bond-

posted, and the FES-UA over-\$50k flag. **Tier scoping:** FES-UA rules report *not applicable* unless the school participates in FES-UA.

The **Review Readiness panel** shows the six sections with color-graded status badges, citations, the \$250k trigger banner, a program-tier selector, and an overall summary with a "**Material → Corrective Action Plan needed**" callout.

3. 2B — Scholarship reconciliation

Addresses AUP **Section IV** at the disbursement-vs-recorded level:

- **Upload** the funding organization's disbursement records (CSV/XLSX, parsed in-browser) with a flexible **column-mapping** step (funding-org headers vary), or add rows manually.
- The engine produces **total disbursed**, **per-program** and **per-month** breakdowns, and reconciles against the recorded scholarship revenue → a **matched / variance / needs-data** verdict (tolerance: the greater of \$1 or 0.5%).
- **Anomaly detection:** duplicates, negative amounts, zero amounts, dates outside the period, unknown program, missing amount.
- A one-click "**use disbursed total as recorded scholarship revenue**" keeps 2A and 2B consistent and feeds the \$250k trigger with real data.

4. 2C — Year-end checklist + workpapers packet

- **Guided checklist:** one procedure item per AUP rule (grouped by section, each with its citation) **plus** a fixed **document-gathering list** (bank statements + reconciliations, DOE approval letter, disbursement records, student subledgers, GL/trial balance, prior-year AUP + CAP, surety bond if applicable). Each item is marked *pending / done / n-a* with notes, and a progress meter distinguishes "done" from "n/a" so a school can't accidentally look ready.
- **Workpapers packet:** a one-click, print-ready document that **aggregates everything** for the reviewer — a cover (school, period, ruleset version, prepared date, disclaimer), the **statements**, the **reconciliation schedule**, the **compliance findings schedule**, and the **Corrective Action Plan**. It reads the saved snapshot — it never re-computes the statement math.

5. 2D — Corrective Action Plan scaffolding

- Every **material** (and, optionally, reportable) finding **auto-scaffolds** a CAP entry pre-filled from a per-rule, AUP-grounded template — e.g. the 60-day-reconciliation finding

pre-fills "Implement a documented monthly bank-reconciliation process completed within 60 days of month-end and independently reviewed..."

- Each entry is **editable** (root cause, corrective action, responsible party, target date) with a **status** (open / in progress / complete) and **persists**.
 - Entries **self-resolve**: when the underlying finding clears (e.g. reconciliation is now within 60 days), the entry is flagged resolved rather than deleted, preserving the record.
 - The CAP **exports** to a clean, print-ready document with the readiness disclaimer.
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Cross-cutting characteristics

- **Security & tenancy.** Every endpoint is role-gated (Owner / Accountant / Viewer), scoped to the school, and tied to an active subscription or live trial (a lapsed account is paused with a friendly prompt, never a raw error). Malformed identifiers are rejected cleanly.
- **Reproducibility.** Metrics, rules, reconciliation, and CAP scaffolding are all pure and deterministic — re-running on the same inputs yields byte-identical results. The frozen financial engine is never modified by these modules.
- **Consistent, on-brand UI.** Both modules share the navy/gold design language, animated-but-tasteful motion (respecting reduced-motion preferences), and a consistent good/watch/risk status palette.
- **Export.** Results print to clean PDF documents (the analytics drawer, the CAP, and the workpapers packet), and the underlying statements export to PDF/Excel as before.

What these modules intentionally leave for later

- **Automatic disbursement pull.** Phase 2B ingests an uploaded funding-org file; the *automatic* connector to the scholarship system (e.g. Step Up For Students) is **Phase 6 — Integrations**.
 - **Per-student bank-deposit tracing.** The deepest AUP §IV procedure (sampling individual ACH payments to bank deposits and student subledgers) remains a **checklist** item, since it needs bank/subledger data the product doesn't yet hold.
 - **Live AI narration.** The AI insight runs on a deterministic generator by default; richer Claude-written narratives activate when an API key is configured.
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This document reflects the state of FinRep as of June 2026: Phases 1, 4, and 2 complete. It is generated from `docs/FEATURES_PHASE_2_AND_4.md` via `scripts/md-to-pdf.py` (run with `python3.10`). The compliance content is a readiness aid grounded in cited Florida statutes

and the published AUP template — it is not the official AUP submission or legal/audit advice.